Saqib Ahmad Zargar

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Research Summary

PhD researcher using **imaging spectroscopy** and radiative transfer to study how **light-absorbing particles**, especially **mineral dust**, darken Himalayan snow and accelerate melt. At **JPL**, I developed a **DISORT-based retrieval of dust in snow from EMIT** spectra, grounded in Himalayan field spectroscopy and snow sampling. I also use in-situ data to constrain and evaluate **SNICAR** and **WRF-Chem** simulations of snow darkening.

Education

PhD, Civil Engineering (Environmental Engineering) Aug 2021 – Present
Indian Institute of Technology Madras *Advisor: Dr. Chandan Sarangi* *Chennai, India*
- Thesis focus: light-absorbing aerosols in Western Himalayan seasonal snow — concentrations, radiative forcing, and hyperspectral satellite detection | CGPA **9.23/10** | Expected defense: Feb 2027

M.Tech, Environmental Engineering Aug 2019 – Jul 2021
Pondicherry University *Puducherry, India*
- Thesis: Optimization of Dal Lake water-quality monitoring using MSTs and ANN | CGPA **9.28/10**

B.Tech, Civil Engineering Aug 2013 – Jul 2017
Islamic University of Science and Technology (IUST) *Pulwama, J&K, India*
- Project: Dynamic isolation of buildings using low-cost bearings | CGPA **8.23/9.5**

Research Experience

Ablation-Season Snowfall, Aerosol Wet Deposition and Snow Darkening in High Mountain Asia 2022 – Present
IIT Madras *Chennai, India*
- Showed from satellite observations (2000–2020) that ablation-season (March–June) **snow-cover duration** and **low-intensity snowfall frequency** increased across the Western Himalaya and Hindu Kush, strongest at mid-elevations; attributed the trend to a deepening upper-tropospheric trough and moisture convergence
- Linked this snowfall enhancement to the seasonal peak in aerosol loading: enhanced **LAP wet deposition** drives rising satellite-derived snow darkening and radiative forcing, showing that more snowfall can **accelerate rather than buffer glacier melt** (first-author; under review)

Speciated Light-Absorbing Particles and Radiative Forcing in Western Himalayan Snow (DRDO Project) Dec 2021 – Dec 2024
IIT Madras *(Dec 2021 – Dec 2024)* *Chennai, India*
- Built a **depth- and time-resolved record of BC, BrC and dust** in surface snow through the pre-monsoon melt season, with concurrent aerosol, deposition-flux and meteorological observations; showed dust-dominated loading governed by episodic westerly transport and precipitation timing
- Quantified LAP radiative forcing with **SNICAR**, validated against spectroradiometer retrievals; showed that **brown carbon** contributes a non-negligible share of absorption and photobleaches as the snowpack ages, so it can be neither omitted nor treated as spectrally static

Remote Retrieval of Dust-on-Snow from NASA EMIT Imaging Spectroscopy Mar 2025 – Feb 2026
NASA Jet Propulsion Laboratory *(Visiting Student Researcher) / IIT Madras* *Pasadena, USA*
- Developed a **DISORT radiative-transfer inversion** to retrieve **dust concentration in snow** from EMIT hyperspectral snow reflectance and albedo products
- Applied the retrieval to Himalayan scenes (Sinthan Pass, Razdan Pass) and a growing High Mountain Asia scene set, with quality assessment of EMIT snow products and their uncertainties

Regional Modeling of LAP Deposition on Himalayan Snow with WRF-Chem Apr 2023 – Present
IIT Madras *Chennai, India*
- Configured WRF-Chem for LAP deposition on snow; traced and fixed a **CLM land-surface bug** causing spurious snow loss over Himalayan grid cells

International Collaborations on Aerosol Optics and Glacier Monitoring 2022 – Present

***PNNL** (USA), **FMI** (Finland), **CryoScope** (Indo-European)*

- **PNNL** (Aug 2022 – Aug 2024): Himalayan aerosol optical properties and composition; complex refractive index of a polluted dust storm
- **FMI** (Aug 2023 – present): brown carbon absorption in snow | **CryoScope** (2024 – present): AWS installation and lake-discharge monitoring in Himalayan glacier catchments

Publications

Peer-reviewed

1. Singh, P., Vijay, S., Banerjee, A., Sarangi, C., Rashid, I., & **Zargar, S. A.** (2025). Quantifying short-term backwasting rates of a supraglacial ice cliff at Machoi Glacier in the Indian Himalaya. *Journal of Glaciology*, 71, e17. doi:10.1017/jog.2024.98
   Contribution: radiation data acquisition and analysis

Under review / In preparation

2. **Zargar, S. A.**, Sarangi, C., Bhariti, P., Deb, P., Banerjee, A., & Rittger, K. Rising ablation-period snowfall darkens the Himalayan snowpack. *Geophysical Research Letters* (under review).
3. Chandel, A. S., Sarangi, C., Nahar, N., **Zargar, S. A.**, et al. (2026). Characterization of the complex refractive index of a polluted dust storm over the Western Himalayas. *EGUsphere* (preprint). doi:10.5194/egusphere-2026-424
4. **Zargar, S. A.**, et al. Speciated light-absorbing particle concentrations, deposition and radiative forcing in Western Himalayan snow. Target: *The Cryosphere* (Final Draft ready).

Conference Presentations

- **Zargar, S. A.**, Sarangi, C., Bhariti, P., Deb, P., Banerjee, A., & Rittger, K. (2026). Evidence of increasing trend of snow cover in Himalayas implicate snow darkening. **EGU General Assembly 2026**, Vienna (**oral**), EGU26-13762. doi:10.5194/egusphere-egu26-13762

Fellowships, Awards and Honors

- **JPL Visiting Student Researcher Program (JVSRP)**, NASA Jet Propulsion Laboratory 2025
- **International Immersion Experience (IIE) Award**, IIT Madras 2025
- HTRA Doctoral Scholarship, Ministry of Education, IIT Madras 2021–2026
- Startup **HailSafe Orchards** selected for incubation at **Nirmaan, IIT Madras** — WRF-based orchard-scale hail forecasting with forecast-triggered automatic hail nets 2026
- UGC NET (**99.7** percentile, 2019); GATE (**91.23** percentile, 2018; **90.45** percentile, 2022); **All-India Rank 2**, Pondicherry University entrance (2019)

Technical Skills

Programming:	Python (NumPy, xarray, SciPy, mpi4py, h5py, netCDF4, Cartopy, GeoPandas), MATLAB; HPC
Models:	WRF, WRF-Chem, SNICAR, DISORT, SBDART*, OPAC*
Satellite & reanalysis:	EMIT (L2 snow products), MODIS, MERRA-2, ERA5, HAR; Google Earth Engine
Field instruments:	ASD spectroradiometer, aethalometer, SP2, OPC, nephelometer, ceilometer, sunphotometer, pyranometer, pyrgeometer, high-volume PM sampler, integrating-sphere spectrometer, LWCC, OC/EC analyser; AWS installation — Himalayan campaigns 2022–2026
Other:	LaTeX*, MS Office, SPSS*, Revit*, Visual Studio,QGIS*

* Elementary proficiency

Teaching and Training

- Teaching Assistant, Environmental Engineering, IIT Madras Jul–Nov 2022; Jan–Jun 2024
- Training: NT-CCAM Workshop on Climate Impacts of Carbonaceous Aerosols, IITM Pune (2023); National Atmospheric Chemistry Seminar Series, IITM Pune (2023); Glacier Studies Workshop, IISER Pune (2022); Technical and Scientific Writing, IIT Madras (2023)